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Experiment No. 10

Title: Digital Modulation Techniques: Generation and Observation of ASK, FSK, and PSK Signals

Aim

To design and simulate Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), and Binary Phase Shift Keying (BPSK) circuits in LTspice using a PULSE voltage source as the binary message signal and Behavioral Voltage Sources (BV) as modulators. To observe the modulated waveforms in the time domain and verify the modulation behaviour for each technique.

Software Used

LTspice 26.0.1 (High-performance SPICE simulation software).

Theory

1. What is Digital Modulation?

Digital modulation is the process of converting digital data (binary bits 0 and 1) into an analog signal by modifying a property of a high-frequency carrier wave. The carrier signal is usually a sinusoidal wave that can travel efficiently through communication channels such as cables, optical fibers, or wireless media.

In digital modulation, one of the three parameters of the carrier signal is varied according to the digital message signal:

- Amplitude
- Frequency
- Phase

These variations allow the binary data to be transmitted over long distances with improved reliability and bandwidth efficiency.

Digital modulation is widely used in modern communication systems such as mobile communication, Wi-Fi, satellite communication, and digital broadcasting.

2. What is Amplitude Shift Keying (ASK)?

Amplitude Shift Keying (ASK) is a digital modulation technique in which the amplitude of the carrier signal is varied according to the binary data, while the frequency and phase remain constant.

Operation:

- Binary 1 → Carrier signal with amplitude Ac is transmitted

- Binary 0 → Carrier signal amplitude becomes zero (no carrier)

ASK signal equation:

$s(t) = m(t) \times Ac \times \sin(2\pi fct)$

Where

m(t) = binary message signal (0 or 1)

Ac = amplitude of carrier signal fc =

carrier frequency

Thus the amplitude of the carrier switches between two levels based on the input data.

Advantages:

- Simple to generate and detect

Disadvantages:

- Highly sensitive to noise and interference

3. What is Frequency Shift Keying (FSK)?

Frequency Shift Keying (FSK) is a digital modulation technique in which the frequency of the carrier signal is varied according to the binary data, while the amplitude remains constant.

In Binary FSK (BFSK):

- Binary 1 → Carrier frequency = f1 (Mark frequency)
- Binary 0 → Carrier frequency = f2 (Space frequency)

FSK signal equation:

For binary 1 $s(t) =$

$Ac \sin(2\pi f1t)$

For binary 0 $s(t) =$

$Ac \sin(2\pi f2t)$

Where

Ac = carrier amplitude

f1 and f2 = two different carrier frequencies

FSK is more resistant to noise than ASK and is commonly used in radio communication, telemetry systems, and modems.

4. What is Binary Phase Shift Keying (BPSK)?

Binary Phase Shift Keying (BPSK) is a digital modulation technique in which the phase of the carrier signal is changed according to the binary input data, while amplitude and frequency remain constant.

Operation:

- Binary 1 → Carrier phase = 0 degrees
- Binary 0 → Carrier phase = 180 degrees

BPSK signal equation:

$s(t) = Ac \sin(2\pi fct)$ for binary 1 s(t)

$= -Ac \sin(2\pi fct)$ for binary 0

Where

Ac = carrier amplitude fc

= carrier frequency

Thus the carrier signal reverses its phase by 180 degrees when the input bit changes.

Advantages:

- Better noise immunity
- Efficient for long distance communication

Applications:

- Satellite communication
- GPS systems
- Wireless communication

5. Significance of Modulation Index / Deviation Ratio / Modulation Order in Digital Modulation Schemes

Modulation Index:

Modulation index indicates the extent of variation in the carrier signal due to the modulating signal. It determines the strength and quality of modulation.

Deviation Ratio:

Deviation ratio is used mainly in frequency modulation techniques and is defined as

Deviation Ratio = Frequency Deviation / Maximum Modulating Frequency

It indicates how much the carrier frequency deviates from its original value.

Modulation Order (M):

Modulation order represents the number of different symbols used in a modulation scheme.

Examples:

- BPSK → M = 2
- QPSK → M = 4
- 16-QAM → M = 16

Higher modulation order:

- Increases data rate
- Improves bandwidth efficiency
- Requires better signal quality

Mathematical Representation

1. Amplitude Shift Keying (ASK)

In ASK, the amplitude of the carrier is switched between two levels depending on the binary message bit. For OOK (On-Off Keying), the simplest form of ASK:

$$s_{ASK}(t) = m(t) \times Ac \sin(2\pi fct) = V_{msg} * V_{carrier}$$

where $m(t) \in \{0, 1\}$ is the binary message (PULSE source), Ac = 1 is the normalised carrier amplitude, and fc is the carrier frequency.

- When m(t) = 1 (bit-1): carrier is transmitted.
- When m(t) = 0 (bit-0): output is zero.

2. Frequency Shift Keying (FSK)

In FSK, the frequency of the carrier is switched between two frequencies depending on the binary message bit:

$s_{FSK}(t) = Ac \sin(2\pi f_1 t)$ when $m(t) = 0$ (space frequency) $s_{FSK}(t) = Ac \sin(2\pi f_2 t)$ when $m(t) = 1$ (mark frequency) where $f_1 = 50$ kHz (space, bit-0) and $f_2 = 100$ kHz (mark, bit-1). The BV uses an IF() condition to select between the two SINE sources.

3. Binary Phase Shift Keying (BPSK)

In BPSK, the phase of the carrier is shifted by 180° depending on the binary message bit: $sBPSK(t) = d(t) \times Ac \sin(2\pi fct)$ where $d(t) \in \{-1, +1\}$ is the bipolar version of the binary message. Since the PULSE source produces $m(t) \in \{0, 1\}$, the BV maps it to d(t) using:

$$d(t) = 2 \times m(t) - 1$$

Thus: $d(t) = +1$ when $m(t) = 1$ (bit-1, 0° phase), and $d(t) = -1$ when $m(t) = 0$ (bit-0, 180° phase reversal).

Circuit Description and Component Mapping

Common Components (used in all three circuits)

A single PULSE voltage source (Vmsg) serves as the binary message signal for all three modulations. It produces a 0V/1V square wave at a bit rate of 1 kbps (bit period Tb = 1 ms).

ASK — Component Table

Component	Source Type	Signal Produced	Key Setting
Vc	SINE source	Carrier: $\sin(2\pi \cdot fc \cdot t)$	Amplitude = 1V, Freq = 20k, DC offset = 0
Component	**Source Type**	**Signal Produced**	**Key Setting**
Vmsg	PULSE source	Binary message: $m(t) \in \{0V, 1V\}$	Vinitial=0, Von=1, Trise=10n, Tfall=10n, Ton=0.5m, Tperiod=1m
B1	Behavioral Voltage (BV)	ASK output: V(msg) × V(carrier)	V = V(msg) * V(carrier)
R1	Resistor	50Ω output load	Value = 50

FSK — Component Table

Component	Source Type	Signal Produced	Key Setting
Vf1	SINE source	Space carrier: $\sin(2\pi \cdot f_1 \cdot t)$	Amplitude = 1V, Freq = 10k, DC offset = 0
Vf2	SINE source	Mark carrier: $\sin(2\pi \cdot f_2 \cdot t)$	Amplitude = 1V, Freq = 20k, DC offset = 0
Vmsg	PULSE source	Binary message: $m(t) \in \{0V, 1V\}$	Same as ASK: Ton=0.5m, Tperiod=1m
B1	Behavioral Voltage (BV)	FSK output: selects f1 or f2	V = IF(V(msg)>0.5, V(f2), V(f1))
R1	Resistor	50Ω output load	Value = 50

PSK — Component Table

Component	Source Type	Signal Produced	Key Setting
Vc	SINE source	Carrier: $\sin(2\pi \cdot fc \cdot t)$	Amplitude = 1V, Freq = 20k, DC offset = 0
Vmsg	PULSE source	Binary message: $m(t) \in \{0V, 1V\}$	Same as ASK: Ton=0.5m, Tperiod=1m
Component	**Source Type**	**Signal Produced**	**Key Setting**
B1	Behavioral Voltage (BV)	BPSK output: phase-flipped carrier	V = V(carrier) * (2*V(msg) − 1)
R1	Resistor	50Ω output load	Value = 50

Simulation Procedure

Three separate circuits are built — one each for ASK, FSK, and PSK. The PULSE source Vmsg is identical in all three circuits.

Part A: ASK Circuit

***Step A1:* Place the Carrier Source Vc (SINE, 20 kHz, Vamp = 1)**

- Select voltage source and place it on the schematic.
- Right-click on the voltage source → Advanced → select SINE.
- Enter the following values:

  DC offset (Voffset) = 0

  Amplitude (Vamp) = 1 ← normalised carrier

  Frequency = 20k ← fc = 20 kHz

  Tdelay = 0, Theta = 0, Phi = 0
- Right-click on the label V1 and rename it to Vc.
- Press 'G' → place ground below Vc. Connect the negative (−) terminal of Vc to Ground.
- Press 'W' to draw a wire from the positive (+) terminal of Vc.
- Press 'N' → assign Net Name: type 'carrier' and press OK. Paste it on the free wire.
- Hover over the wire — the bottom status bar must show 'carrier' (not N001 or N002).

***Step A2:* Place the Binary Message Source Vmsg (PULSE, 1 kbps)**

- Select voltage source and place it to the right of Vc.
- Right-click on the voltage source → Advanced → select PULSE.
- Enter the following values:

  Vinitial = 0 ← bit-0 level

  Von = 1 ← bit-1 level

  Trise = 10n, Tfall = 10n ← fast transitions

  Ton = 0.5m ← bit period high = 0.5 ms Tperiod =

  1m ← bit period = 1 ms (bit rate = 1 kbps)
- Right-click on the label and rename it to Vmsg.

- Press ‘G’ → connect the negative (−) terminal of Vmsg to Ground.
- Press ‘W’ → draw a wire from the positive (+) terminal of Vmsg.
- Press ‘N’ → assign Net Name: type 'msg' and press OK. Paste it on the free wire.
- Hover over the wire — the bottom status bar must show 'msg'.

Step A3: Place the Behavioral Voltage Source B1 (ASK Multiplier)

- Press ‘P’ → search bv → select Behavioral Voltage → place it to the right of Vc and Vmsg.
- Right-click B1 → in the Value field type EXACTLY:

  V = V(msg) * V(carrier)

B1 Expression — ASK Verification:

V(msg) = m(t) ∈ {0, 1} ← from PULSE source Vmsg (Step A2)

V(carrier) = sin(2π·fc·t) ← from SINE source Vc with Vamp=1 (Step A1)

B1 output = V(msg) × V(carrier)

= m(t) × sin(2π·fc·t)

= $s_{ASK}(t)$ ✓ matches standard ASK (OOK) equation

Step A4: Place Load Resistor R1 and Complete the ASK Circuit

- Press ‘R’ → place resistor. Right-click → set value to 50.
- Press ‘W’ → connect B1 positive (+) terminal to the top of R1.
- Press ‘N’ → label this wire: ASK_out.

  (Note: include your roll number in the label e.g. ASK_out_26)

- Press ‘G’ → ground B1 negative (−) terminal and the bottom of R1.

Step A5: Set Up and Run Transient Simulation (ASK)

- Click Simulate → Edit Simulation Cmd → Transient tab.

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  Set Stop Time: .tran 5m ← 5 milliseconds = 5 complete message bits at 1 kbps.

  Place this directive on the schematic canvas. Click Run.

Step A6: View ASK Waveforms

- Click on the ASK_out wire → adds V(ASK_out) to the plot.
- Click on the msg wire → adds V(msg) to see the binary message.
- Click on the carrier wire → adds V(carrier) to see the unmodulated carrier.
- Observe: carrier present when msg = 1; carrier absent (zero output) when msg = 0.

Part B: FSK Circuit

Step B1: Place the Space Frequency Source Vf1 (SINE, 10 kHz)

- Select voltage source and place it on the schematic.
- Right-click → Advanced → select SINE.
- Enter the following values:

  Amplitude (Vamp) = 1

  Frequency = 50k ← f1 = 10 kHz (space, bit-0)

  Voffset = 0, Tdelay = 0, Theta = 0, Phi = 0
- Rename the source to Vf1.
- Connect the negative (−) terminal to Ground.
- Assign Net Name 'f1' to the wire from the positive (+) terminal.

Step B2: Place the Mark Frequency Source Vf2 (SINE, 20 kHz)

- Select voltage source and place it to the right of Vf1.
- Right-click → Advanced → select SINE.
- Enter the following values:

  Amplitude (Vamp) = 1

-
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  Frequency = 100k ← f1 = 20 kHz (space, bit-0)

  Voffset = 0, Tdelay = 0, Theta = 0, Phi = 0

  Rename the source to Vf2.

  Connect the negative (−) terminal to Ground.

- Assign Net Name 'f2' to the wire from the positive (+) terminal.

Step B3: Place the Binary Message Source Vmsg (PULSE, 1 kbps)

- Repeat Step A2 exactly. Use the same PULSE settings.
- Assign Net Name 'msg' to the wire from the positive (+) terminal.

Step B4: Place the Behavioral Voltage Source B1 (FSK Selector) •

  Press 'P' → search bv → select Behavioral Voltage → place it.

- Right-click B1 → in the Value field type EXACTLY:

  V = IF(V(msg)>0.5, V(f2), V(f1))

B1 Expression — FSK Verification:
V(msg) > 0.5 → bit-1 detected → B1 outputs V(f2) = sin(2π·20k·t) (mark)
V(msg) ≤ 0.5 → bit-0 detected → B1 outputs V(f1) = sin(2π·10k·t) (space)
= $s_{FSK}(t)$ ✓ matches standard FSK equation

Step B5: Place Load Resistor R1 and Complete the FSK Circuit

- Press 'R' → place resistor with value 50.
- Connect B1 positive (+) terminal to the top of R1.
- Assign Net Name 'FSK_out' to this wire.

  (Note: include your roll number in the label e.g. FSK_out_26)

- Ground B1 negative (−) terminal and the bottom of R1.

Step B6: Set Up and Run Transient Simulation (FSK)

- Click Simulate → Edit Simulation Cmd → Transient tab.

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- Set Stop Time: .tran 5m ← 5 ms = 5 complete bits at 1 kbps.
- Place the directive on the schematic canvas. Click Run.

Step B7: View FSK Waveforms

- Click on the FSK_out wire → adds V(FSK_out) to the plot.
- Click on the msg wire → adds V(msg) to see the binary message.

  Click on the f1 wire → adds V(f1) to see the 10 kHz space carrier.

  Click on the f2 wire → adds V(f2) to see the 20 kHz mark carrier.
- Observe: lower frequency (10 kHz) during bit-0; higher frequency (20 kHz) during bit-1.

Part C: BPSK Circuit

Step C1: Place the Carrier Source Vc (SINE, 10 kHz, Vamp = 1)

- Repeat Step A1 with Freq = 10 kHz.
- Assign Net Name 'carrier' to the wire from the positive (+) terminal.

Step C2: Place the Binary Message Source Vmsg (PULSE, 1 kbps)

- Repeat Step A2 exactly.
- Assign Net Name 'msg' to the wire from the positive (+) terminal.

Step C3: Place the Behavioral Voltage Source B1 (BPSK Phase Mapper)

- Press 'P' → search bv → select Behavioral Voltage → place it.
- Right-click B1 → in the Value field type EXACTLY:

  V = V(carrier) * (2*V(msg) - 1)

B1 Expression — BPSK Verification:

When V(msg) = 1 (bit-1): $2\times1 - 1 = +1$ → B1 = +1 × V(carrier) → 0° phase

When V(msg) = 0 (bit-0): $2\times0 - 1 = -1$ → B1 = −1 × V(carrier) → 180° phase reversal

B1 output = $(2\times m(t) - 1) \times \sin(2\pi\cdot fc\cdot t)$

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= $s_{BPSK}(t)$ ✓ matches standard BPSK equation

***Step C4:* Place Load Resistor R1 and Complete the BPSK Circuit**

- Press 'R' → place resistor with value 50.
- Connect B1 positive (+) terminal to the top of R1.
- Assign Net Name 'PSK_out' to this wire.
- Ground B1 negative (−) terminal and the bottom of R1.

***Step C5:* Set Up and Run Transient Simulation (BPSK)**

- Click Simulate → Edit Simulation Cmd → Transient tab.

  Set Stop Time: .tran 5m ← 5 ms = 5 complete bits at 1 kbps.

  Place the directive on the schematic canvas. Click Run.

***Step C6:* View BPSK Waveforms**

- Click on the PSK_out wire → adds V(PSK_out) to the plot.

  (Note: include your roll number in the label e.g. PSK_out_26)

- Click on the msg wire → adds V(msg) to see the binary message.
- Click on the carrier wire → adds V(carrier) to see the unmodulated carrier.
- Observe: carrier phase reversal (180° shift) at every 0→1 or 1→0 transition in the message.

Observations

ASK(Part A) :

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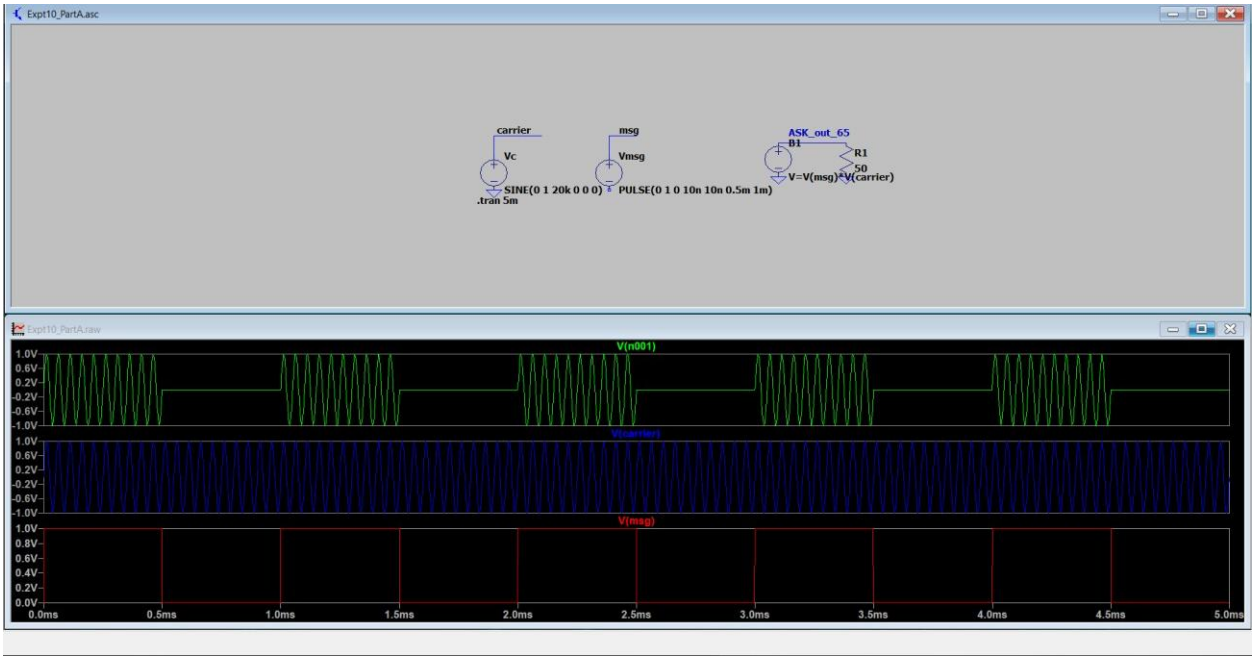
Expt10_PartA.asc
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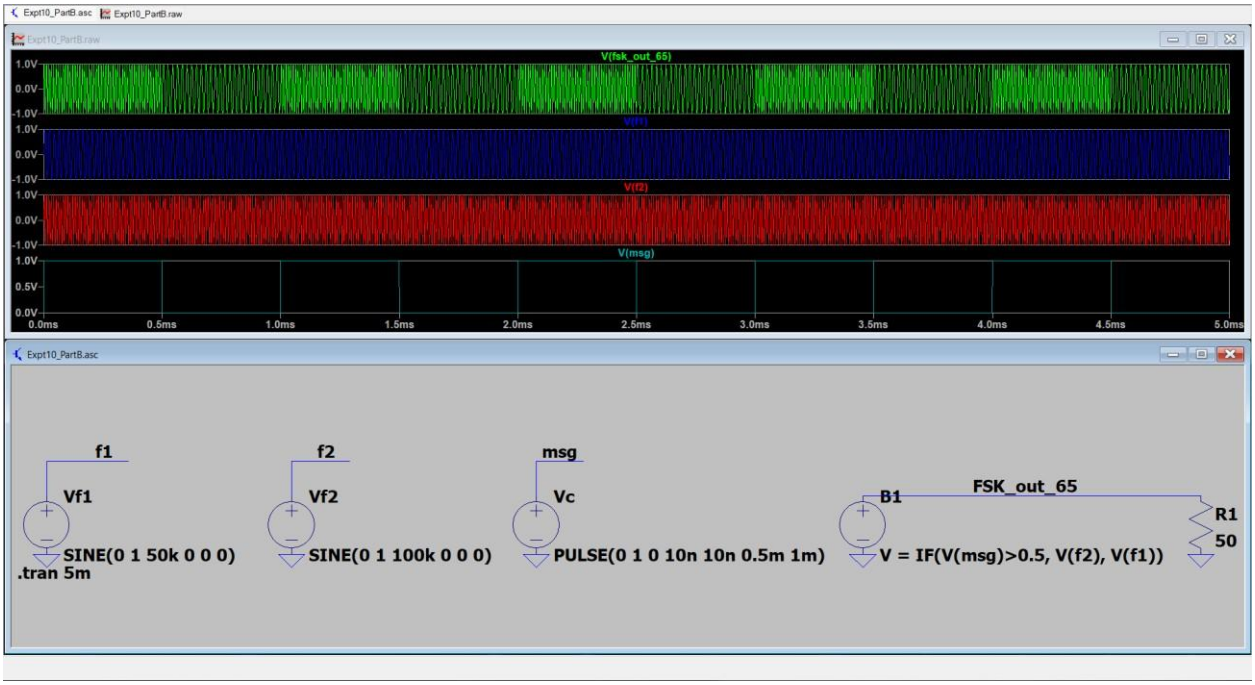
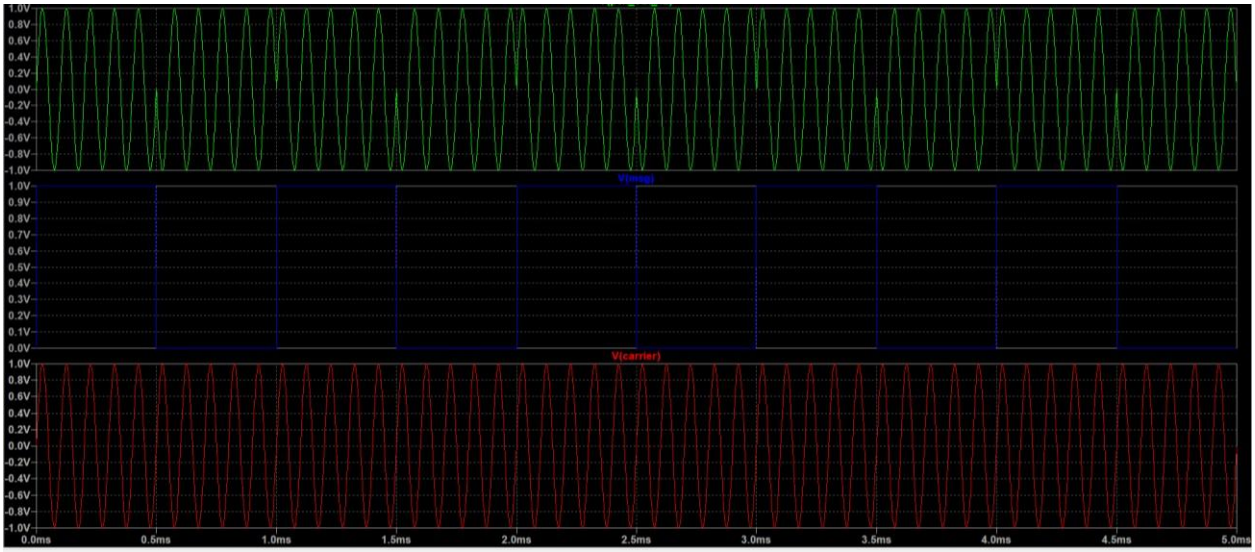
Vc
SINE(0 1 20k 0 0 0)
.tran 5m
msg
Vmsg
PULSE(0 1 0 10n 10n 0.5m 1m)
ASK_out_65
B1
V=V(msg)*V(carrier)
R1
50
Expt10_PartA.raw
V(n001)
V(carrier)
V(msg)
1.0V
0.6V
0.2V
-0.2V
-0.6V
-1.0V
0.8V
0.4V
0.0V
0.0ms
0.5ms
1.0ms
1.5ms
2.0ms
2.5ms
3.0ms
3.5ms
4.0ms
4.5ms
5.0ms

FSK(Part B) :



BPSK(Part C) :



Modulation	Message Bit Pattern	Carrier Freq. / Freqs.	Bit Rate	Observable Feature	Observation (describe waveform)	Verified?
ASK	0V / 1V PULSE	fc = 20 kHz	1 kbps	Carrier on/off with bit	When the message signal is 1, the carrier waveform appears at the output. When the message signal is 0, the output becomes zero (no carrier). The amplitude of the carrier is controlled by the message signal.	Yes

FSK	0V / 1V PULSE	f1=10k, f2=20k	1 kbps	Frequency switches with bit	When the message signal is 0, the output carrier has lower frequency (f1). When the message signal is 1, the output switches to higher frequency (f2). The amplitude remains constant but the frequency changes according to the input bit.	Yes
Modulation	**Message Bit Pattern**	**Carrier Freq. / Freqs.**	**Bit Rate**	**Observable Feature**	**Observation (describe waveform)**	**Verified?**
BPSK	0V / 1V PULSE	fc = 10 kHz	1 kbps	180° phase reversal at transitions	The output carrier maintains the same frequency and amplitude, but the phase reverses by 180° whenever the message bit changes. This phase inversion is visible at the transitions of the message signal.	Yes

Conclusion

Three digital modulation schemes were simulated in LTspice using a common PULSE voltage source (Vmsg, 0V/1V, 1 kbps) and Behavioral Voltage Sources (BV) as modulators.

- **ASK (OOK):** The BV expression V = V(msg) × V(carrier) directly implements On-Off Keying. The 20 kHz carrier was present when the message bit was 1 and absent when the bit was 0. The modulation was confirmed from the V(ASK_out) waveform.
- **FSK:** The BV expression V = IF(V(msg)>0.5, V(f2), V(f1)) switched between the 10 kHz space carrier (bit-0) and the 20 kHz mark carrier (bit-1). The frequency transition was clearly visible at each bit boundary in the V(FSK_out) waveform.
- **BPSK:** The BV expression V = V(carrier) × (2×V(msg) − 1) mapped bit-1 to +1 (0° phase) and bit-0 to −1 (180° phase reversal). The phase inversion at each 0↔1 and 1↔0 transition was clearly observable in the V(PSK_out) waveform.

All three circuits confirm the fundamental principle of digital modulation: a property of the carrier (amplitude, frequency, or phase) is varied in accordance with the binary message data.